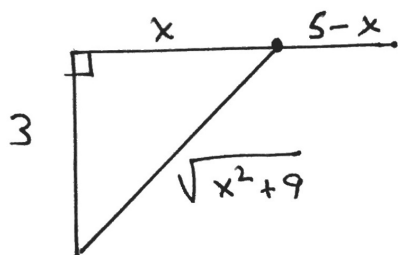


COURSE PACK SOLUTIONS - PAGE 9

①

1.



$$\text{Total Time} = T_{\text{row}} + T_{\text{walk}}$$

$$= \frac{\sqrt{x^2 + 9}}{2} + \frac{5-x}{4}$$

$$(\text{Total Time})' = \frac{2x}{4\sqrt{x^2 + 9}} - \frac{1}{4}$$

$$\frac{x}{2\sqrt{x^2 + 9}} - \frac{1}{4} = 0$$

$$\frac{x}{\sqrt{x^2 + 9}} = \frac{1}{2}$$

$$2x = \sqrt{x^2 + 9}$$

$$4x^2 = x^2 + 9$$

$$3x^2 = 9$$

$$x^2 = 3$$

$$x = \sqrt{3}$$

$$x \doteq 1.73 \text{ km.}$$

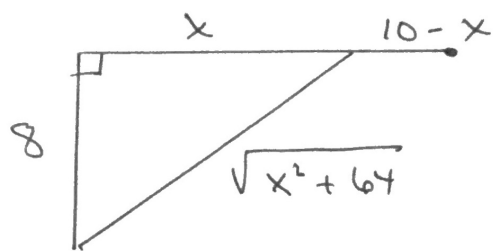
$\therefore$ He should land

3.27 km (5 - 1.73)

from the desired
destination.

(2)

2.



$$a) \text{ Total Time} = T_{\text{row}} + T_{\text{run}}$$

$$= \frac{\sqrt{x^2 + 64}}{4} + \frac{10-x}{6}$$

$$\left(\text{Total Time} \right)' = \frac{x}{4\sqrt{x^2 + 64}} - \frac{1}{6}$$

$$\frac{x}{4\sqrt{x^2 + 64}} - \frac{1}{6} = 0$$

$$\frac{x}{\sqrt{x^2 + 64}} = \frac{2}{3}$$

$$9x^2 = 4(x^2 + 64)$$

$$5x^2 = 256$$

$$x \approx 7.12$$

$\therefore$ Land at 2.84 km from desired point.

$$\therefore T(7.12) = 3.15 \text{ hours.}$$

(3)

2b.

$$\text{Total Time} = \frac{\sqrt{x^2 + 64}}{4} + \frac{10 - x}{5}$$

$$\left(\frac{\text{Total}}{\text{Time}} \right)' = \frac{x}{4\sqrt{x^2 + 64}} - \frac{1}{5}$$

$$\frac{x}{4\sqrt{x^2 + 64}} - \frac{1}{5} = 0$$

$$\frac{x}{4\sqrt{x^2 + 64}} = \frac{1}{5}$$

$$25x^2 = 16x^2 + 1024$$

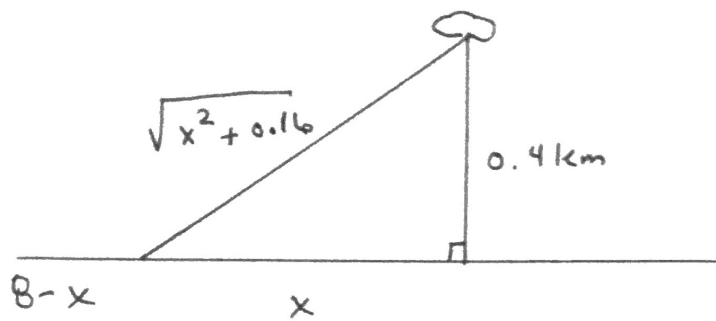
$$9x^2 = 1024$$

$$x \doteq 10.67 > 10 \text{ km}$$

$\therefore$ row all the way!

3.

4



$$\text{Total Time} = T_{\text{swim}} + T_{\text{walk}}$$

$$= \frac{\sqrt{x^2 + 0.16}}{3} + \frac{8-x}{5}$$

$$\left(\text{Total Time} \right)' = \frac{x}{3\sqrt{x^2 + 0.16}} - \frac{1}{5}$$

$$\frac{x}{3\sqrt{x^2 + 0.16}} - \frac{1}{5} = 0$$

$$\frac{x}{3\sqrt{x^2 + 0.16}} = \frac{1}{5}$$

$$25x^2 = 9x^2 + 1.44$$

$$x^2 = 0.09$$

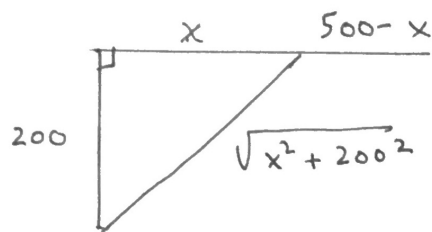
$$x = 0.3$$

$$\therefore \text{Walk} = 0.8 - 0.3$$

walk = 0.5 km & swim the rest.

(5)

4.



$$\text{Total Time} = T_{\text{row}} + T_{\text{walk}}$$

$$= \frac{\sqrt{x^2 + 200^2}}{3} + \frac{500-x}{5}$$

$$\left(\text{Total Time} \right)' = \frac{x}{3\sqrt{x^2 + 200^2}} - \frac{1}{5}$$

$$\frac{x}{3\sqrt{x^2 + 200^2}} - \frac{1}{5} = 0$$

$$\frac{x}{3\sqrt{x^2 + 200^2}} = \frac{1}{5}$$

$$25x^2 = 9x^2 + 360,000$$

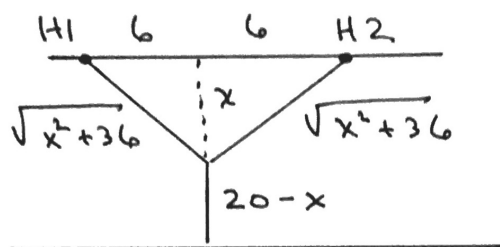
$$16x^2 = 360,000$$

$$x = 150$$

$$\therefore T(150) \doteq 153.3 \text{ seconds.}$$

6

5.



$$\text{Length} = 2\sqrt{x^2 + 36} + 20 - x$$

$$(\text{Length})' = \frac{2x}{\sqrt{x^2 + 36}} - 1$$

$$\frac{2x}{\sqrt{x^2 + 36}} - 1 = 0$$

$$\frac{2x}{\sqrt{x^2 + 36}} = 1$$

$$4x^2 = x^2 + 36$$

$$3x^2 = 36$$

$$x^2 = 12$$

$$x \doteq 3.46$$

$$\Rightarrow 20 - x$$

$$= 16.54 \text{ km}$$

$\therefore$ The junction box should be placed 16.54 km from the highway.